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Forms of presentism in the history of science. Rethinking the project of historical epistemology

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ABSTRACT

Since the late 1980s, presentism has seen a resurgence among some historians of science. Most of them draw a line between a good form of presentism and typical anachronism, but where the line should be drawn remains an open question. The present article aims at resolving this problem. In the first part I define the four main distinct forms of presentism at work in the history of science and the different purposes they serve. Based on this typology, the second part reconsiders what counts as anachronism, Whiggism and positivist history. This clarification is used as a basis to rethink the research program of historical epistemology in the third section. Throughout this article, I examine the conceptual core of historical epistemology more than its actual history, from Bachelard to Foucault or others. Its project should be defined – as Canguilhem suggested – as an attempt to account for both the contingency and the rationality of science. As such, historical epistemology is based on a complex fifth form of presentism, which I call *critical presentism*. The critical relation at stake not only works from the present to the past, because of the acknowledged rationality of science, but also from the past to the present because of the contingency and historicity of scientific knowledge.

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1. Introduction

Accusations of "Whiggism" have long been a staple of debate among historians of science. Much has been written about the subject since the mid-twentieth century, when history of science became a professional practice, gaining a modicum of independence from both science and philosophy (Hall, 1983; Harrison, 1987; Wilson & Ashplant, 1988). In order to assert their new status and emphasize the difference between them and their predecessors, the first professional historians of science chose to strongly reject all forms of retrospective analysis in their field. Therefore, from the 1960s to the 1980s, "whiggish" became a synonym of incompetent and the worst insult that could be levelled against a historian of science. Yet, as has been noted, in the 1980s things began to turn around, and several historians (who were not scientists to begin with) argued that the practice of history could not do without at least a mild form of "present-centeredness" (Rée, 1991; Wilson & Ashplant, 1988). Over the past few years, this trend towards the reconsideration of presentism in history of science has gained momentum, and more and more historians deem "certain forms" of presentism and anachronism necessary (Alvargonzàlez, 2013; Jardine, 2000; Oreskes, 2013; Tosh, 2003). More generally, the issues of why and how historians of science conduct their research and produce narratives of the past have been raised again, giving further credence to the presentist approach. For example, Oren Harman and Alexandre Métraux edited in December 2013 an entire issue of *Science in Context* devoted to the topic, entitled "Approaches, styles, and narratives: reflections on doing history of science" (Harman and Métraux, 2013). All seven contributions in that volume raise the crucial problem for historians of science of the inescapable tension between the present and the past.

This resurgence of presentism in the field of history of science is evidence of its centrality. If one accepts that the objective of science is to produce true explanations, it seems difficult to dismiss all use of present knowledge to trace back and understand its history. This is why, as David Alvargonzàles claims, "history of science is essentially whiggish" (Alvargonzàlez, 2013) – at least to some extent. The question is therefore no longer *if* we have to make room for presentism, but rather *how* we should use presentism. Throughout this paper I use the term "presentism" in an inclusive sense, referring to any form of retrospective analysis that makes some room for present day science. This definition implies no value judgement: presentism can lead to misinterpreting the past just as it can also help better understand it. The aim of this article is to bring insights into the pivotal question of presentism in two related

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ways: by clarifying the terms and concepts at stake (Sections 2 and 3) and by characterizing the project of historical epistemology as a way to solve the issue of presentism in the history of science (Section 4).

First, in light of much of the recent (and less recent) literature on that topic, I propose a classification of four different types of presentism, from the less controversial in practice to the most disputed. I name these forms *empirical*, *descriptive*, *causal-narrative* and *normative* presentisms. I examine potential ways of "using them without any abuse" (Jardine, 2000) by emphasizing the practical conditions under which these types of presentism may be legitimate. I also argue that each of these forms of presentism may yield distinct and important contributions to our understanding of the past.

On the basis of this typology, the second part of the article proposes definitions of anachronism on the one hand and Whiggism and positivist history on the other (the latter two categories are treated as almost synonymous here, as most historians use them interchangeably). The above categories apply to retrospective analyses in which three of the four forms of presentism are not used consistently. This characterization highlights that Whiggism has two close but nonetheless different meanings, which are very often conflated and ill-defined. The first consists in a misuse of *causal-narrative presentism* (Whiggism as history necessarily leading to present science, in a finalistic approach), whereas the second is a misuse of *normative presentism* (Whiggism as history in which past science is reduced to an incomplete version of present science).

In the last part of the article, I reconsider the very nature of historical epistemology in close connection with the question of presentism. The term "historical epistemology" has at least two acceptations. It can refer to a tradition in history and philosophy of science, especially in France, running from Bachelard to Canguilhem, Foucault and others (Braunstein, 2002, 2006). Yet it can also be understood as a method, not necessarily linked to the French tradition (Braunstein, 2008b). It is this more normative approach that I am primarily interested in: regardless of the concept's history, my aim is to show that historical epistemology is best characterized as a way not only to raise the problem of presentism in the history of science but to *solve* it. I argue that historical epistemology accounts for both the historicity and the rationality of science and thereby escapes the dead ends of Whiggism and positivism on the one hand and relativism on the other. Despite its normative focus, this section also draws on aspects of Georges Canguilhem's work, which has received increasing scrutiny in recent years (Chimisso, 2015; Méthot, 2013). Unlike Bachelard, Canguilhem gave close consideration to historicity and contingency in history of science. Unlike Foucault, he remained convinced of the necessity to use the present as a yardstick.

2. Four forms of presentism

In one of the most recent articles on presentism, David Alvargonzàlez makes a very convincing case for a "certain conceptual anachronism" in the field of history of science (Alvargonzàlez, 2013, p. 95). Others have also supported this claim in the past few years, particularly since Nick Jardine's influential 2000 paper "Uses and abuses of anachronism in the history of science" (Jardine, 2000). Since then, it has generally been acknowledged that the practice of history of science needs to refer to the present for a variety of reasons. Most historians of science make a distinction between a "good" and a "bad" form of presentism: the former helps us understand the past while the second only distorts it under the weight of the present.

For example, Nick Jardine proposed in his 2000 paper a distinction between a "legitimate" and a "vicious" form of anachronism. More recently, in 2009, Oscar Moro-Abadia devoted a whole article to this question, and, following Jardine, tried to establish a clear separation between "present-centeredness" and "anachronistic history" (Moro-Abadia, 2009). Most of the literature on that topic is based on such dichotomies. My aim in this section is to show that we must now go beyond a dichotomic understanding of the nature of presentism: the question is not only about professional ethics and a choice between good and bad; we should be concerned with providing explicit definitions of the different forms of presentism applied in the field of history of science.

These forms have been more or less clearly acknowledged since the late 1970s. For instance, in 1979, David Hull published a defense of presentism in which he already considered three different forms of presentism:

In this paper, I discuss three sorts of presentism, reading present-day meanings, principle of reasoning, and empirical knowledge back into earlier periods. I argue that in all three cases knowledge of present-day language, logic, and science is necessary not only for investigating the past but also for communicating the results of these investigations to the historian's contemporaries (Hull, 1979, p. 4).

My goal here is to develop clear definitions of what I consider to be the main forms of presentism on the basis of the existing literature and of my own experience as a historian of biology. Three of these forms could be used both in a "bad" and in a "good" way depending on the competence of the historian. Therefore the first question that must be answered is not about the legitimacy of presentism as a whole, but about its various definitions and functions in the construction of historical narratives. In this section, I introduce a distinction between the four following types of presentism: *empirical presentism, descriptive presentism*, *causal-narrative presentism* and *normative presentism*. I consider these forms of presentism as mainly exclusive: each of them defines a specific way of using the present in order to understand the past which, in the main, does not overlap with the others.

2.1. Empirical presentism

In 1983, French historian Guy Beaujouan already regretted that historians of science – and especially historians of ancient science – almost systematically neglected the "operational substratum" of knowledge and did not "look for the realities that might underpin the natural wonders reported in chronicles"[1] (Beaujouan, 1983, pp. 351–352). In his view history of science all too often failed to consider what present science could tell us about the materiality of natural phenomena. This *empirical presentism* is extensively discussed in most of the articles devoted to Whiggism and anachronism. It could be defined as *the use of present-day knowledge to help specify the characteristics of the empirical substratum based on which past interpretations were made, regardless of the content of these interpretations*.

Much of Hull (1979) paper refers to this form of presentism. More recently, Nick Jardine and Nick Tosh also defended this kind of practice, and both of them strongly oppose what they consider as being a complete sterilization of intellectual history because of an extremist understanding of anti-presentist methodology:

All too often recent historians of science have abandoned common sense in their flight from presentism. Since it is presentist to appeal to knowledge we possess but that the subjects of our historical studies did not, we are supposedly not free to deploy in our interpretations and explanations such facts as, for example, our diagnoses of the diseases afflicting past persons, or our computed

[1] My translation.

estimates of past coordinates in the night sky (Jardine, 2003, p. 134).

What Jardine, Tosh and others have in mind is not in doubt. They are dismayed by provocative statements made by people like Bruno Latour, for whom the anthropologist-historian is absolutely not allowed to escape the conceptual world of the actors he is studying (Latour, 1996, pp. 247–249). Yet this extreme form of anti-presentism seems to me rather uncommon in the work of contemporary historians, with a few exceptions including Latour. One can indeed hardly imagine a rational argument supporting the claim that we should not take advantage of present-day technical and scientific advances to improve our understanding of the past world. This very special form of presentism does not pose any risk of distortion because it has no bearing on our interpretation of history.

For example, it is essential to have in mind present-day knowledge about the very specific chromosomal properties of the plant *Oenothera lamarckiana* in order to better evaluate the mutation theory developed by Hugo de Vries in the early twentieth century (De Vries, 1901-1903). Knowing that this species has (and had 100 years ago) a very original ability to produce massive chromosomic changes is fundamental to appreciating the impact of the model-organism chosen by de Vries on the general shape of his evolutionary explanations. Of course, this does not mean at all that it is not worth placing de Vries's science in the context of the ideas of his time concerning individual variation. The only thing at stake here is that this kind of present-day knowledge represents a positive contribution for the understanding of the past. In this sense I concur with Nick Jardine's dismissal of the stances taken by Latour and others as "sillinesses" (Jardine, 2004, p. 270).

Empirical presentism is helpful in history of science to retrospectively understand the constraints of the intrinsic structure of the natural world on past scientific thoughts and explanations. The only pre-requisites to be able to use this form of presentism – precisely those denied by Latour and the most radical anti-realist sociologists of knowledge – are that:

(1) The natural world exists regardless of what we know about it.
(2) The natural world now is to a very large extent the same as the one studied by scientists in the past (in the words of Geoffrey Lloyd, "natural phenomena […] manifest […] impressive and robust stabilities" (Lloyd, 2013, p. 591)).

The very possibility of science requires accepting the first postulate. Paul Boghossian has convincingly shown the impossibility of a radical constructivist approach, and I refer the reader to his book for the detailed demonstration (Boghossian, 2006). Therefore, historians of science have to accept that the science of the past was a discourse about *something*, and that it helps to know as much as possible about it.

The second postulate must be understood in two ways. Most fundamentally, it implies that natural laws and regularities are constant. Henri Poincaré showed a long time ago (Poincaré, 2004 [1911]) that this is a condition for science to exist (which brings us back to the first postulate). But in the biological sciences, this postulate has an empirical dimension that must not be taken for granted. It implies that biological objects (species for instance) remain the same on the scale of human history: what scientists of the past were studying is not essentially different from what present scientists are examining. Given what we know about biological evolution, this hypothesis is reasonable. For instance, there is no good reason to think that the *Oenothera lamarckiana* described by Hugo de Vries is not the same as the species we know today.

Thus, if one accepts that "the world out there" exists there is no reason not to take into account what we know today about it in order to improve our understanding of past theories and past explanations. This *empirical presentism* does not create any bias and must accordingly be used systematically, without restriction.

2.2. Descriptive presentism

In his 1979 paper, David Hull insisted that the function of the historian of science is to make past explanations and practices understandable to present readers (Hull, 1979, pp. 14–15). This didactic constraint of communicating in the present makes a wholesale rejection of presentism impossible. In this sense at least, historians cannot escape presentism. The historian immersed in the period he is working on has to accept leaving that period in order to tell something about it from the present (Joly, 2013, p. 10): he must describe the past in terms that are understandable in the present. I call this form of presentism *descriptive presentism*, and define it as follows: *the comparison/transcription/translation of the structure of a past explanation in terms that are understandable in the present*.

This is a particularly sensitive issue in the case of past theories and past explanations which were historically complete dead ends, such as the phlogiston theory in chemistry or Aristotelian physics. In such cases, historians directly face the problem of translation because they have to render understandable past concepts that are no longer alive in present knowledge. In this form of presentism, the translation must be done both at the level of concepts as individual entities, and at the level of the complex net of interactions in which they are engaged.

In my view, there is no theoretical argument that would legitimate this kind of presentism in all types of situations. On the contrary, historians who refuse to distance themselves from past formulations and discourses, like Quentin Skinner, usually justify their position by using the Kuhnian concept of incommensurability (Skinner, 1969, p. 7). If one adopts a strictly Kuhnian perspective, then it seems that any attempt of translation or transcription is impossible, and one can either stick to what the actors said or misconstrue their thought. Yet the fact remains that until now, *descriptive presentism* has proven to be efficient, useful and necessary even in the most difficult cases such as Aristotelian physics. We (i.e., post-Newtonian scholars) are able to understand both Aristotle's science and ours and to *contrast* them. This is precisely what Thomas Kuhn did in his seminal book *The Structure of Scientific Revolutions* and this is why it remains fascinating more than half a century after it was first published (Kuhn, 1962).

2.3. Causal-narrative presentism

Historians of science do not only have to be understandable; they also have to produce narratives, i.e. causal sequences with a beginning and an end. As Nick Tosh has already stressed, this is the consequence of an irreducible difference between anthropology and history:

However, there is an important and obvious asymmetry between history and anthropology: time has an intrinsic direction; geographical or cultural separation does not. Even the most hard-line methodological relativist on anthropological questions would be hard pressed to deny the historian the privilege of retrospection: to put it crudely, in history *we know what happened afterwards, and the actors don't* (Tosh, 2003, p. 653, emphasis in the original).

Following Tosh, *narrative presentism* could be defined as *the use of events that occurred after the period considered to identify causal historical processes and to give perspective to the narrative produced*.

Firstly, the existence of causal series in history directly provides historians with a criterion for the selection of past events. They are

preferentially interested in the events of the past which subsequently proved to be important. This is what Naomi Oreskes recently labelled "motivational presentism" (Oreskes, 2013): the fact that historians define their object in relation to what they consider as being relevant to current issues. For example, it makes sense to work on Darwin's ideas because of the far-reaching influence of these ideas in many fields since 1859. *Narrative presentism* is indeed perfectly legitimate not because the present is the necessary goal of the past, but because the past effectively and causally produces the present.

Moreover, a distinct form of *narrative presentism* has recently been revived due to the growing influence of counterfactual history or virtual history. Historians and then historians of science have been investigating the possibility of rewriting history by conducting thought-experiments that always follow the same precise structure: what would happen to event Y if earlier event X had been Z instead (Radick, 2008)? For instance, the historian of evolutionary thought Peter Bowler published a whole book based on this methodology: *Darwin Deleted: Imagining a World without Darwin* (Bowler, 2013). The purpose of virtual history is to test historical causality by using our knowledge of later events. Counterfactual history is thus necessarily presentist. This form of *causal-narrative presentism*, based on the knowledge of later situations, is used in order to identify in the past the potential sources of causal events that led to these situations. The construction of stories and the recent surge of interest in the narrative form have forced historians to attempt to identify causal processes in the course of history, and retrospective analysis is the main way of achieving this goal. Niall Ferguson clearly emphasizes this crucial aspect of counterfactual history when he writes that "if we want to say anything about causation in the past without invoking covering laws, we really have to use counterfactuals, if only to test our causal hypotheses" (Ferguson, 1999, p. 81). In this respect, counterfactual history must be seen as the most radical and promising form of *causal-narrative presentism*.

Coming from a completely different perspective, Andrew Cunningham published in 1988 an influential and programmatic article entitled "Getting the game right: Some plain words on the identity and invention of science" in which he proposed a very strong anti-presentist methodology (focusing especially on what I call here *causal-narrative presentism*). Following the thesis proposed by Cunningham in 1988 requires limiting one's work to human activities that were categorized under the same scientific label in the past (e.g., the history of biology should only take into consideration what was considered to be biology in the past regardless of the current definition of the discipline). Cunningham emphasized this methodological principle when he wrote that "to describe [past people] as having been engaged in science can only be a true description of their activity if it could have been *their own* description of their activity" (Cunningham, 1988, p. 378). In other words, this methodological positioning necessarily implies that the past of a given science is reducible to this same science in the past. This specific issue was independently outlined by Canguilhem in 1976, in a famous essay entitled "The Role of Epistemology in Contemporary History of Science":

The pure history of eighteenth century botany can include in the name of botany nothing more than what botanists at the time assigned themselves as their domain of exploration. Pure history reduces the science it studies to the field of investigation assigned to it by past scholars and to their outlook on that field. But is this science of the past a past of today's science? (Canguilhem, 2000 [1976], p. 13).[2]

For Canguilhem, the answer to this question was clearly no. Scientific disciplines have historically emerged from the junction of heterogeneous lines of thought, some of them very different from the final outcome. For example, Canguilhem argued that taking into account the corpuscular theories of matter and light by Sir Isaac Newton is fundamental in order to understand the history of cell theory in the nineteenth century (Canguilhem 2003 [1946], pp. 68–70). Newton was important not because he described himself as a cellular biologist but because his very influential ideas causally initiated what Daniel J. Nicholson calls "biological atomism" (Nicholson, 2010).

2.4. Normative presentism

In contrast to the first three forms of presentism, the fourth is – surprisingly – less discussed and not always taken into account in the literature, at least by historians who were not initially scientists. As I see it, however, it is the most important one for historians of science and the only one that is exclusive to history of science. Rupert Hall dramatically expressed what is at stake in this sort of presentism when he stressed the difference between history and history of science:

The modern historian cannot but be oppressed by the consciousness of error, like an advocate in court who knows that his client is guilty. True, historians are also conscious of the errors of Burgoyne, Villeneuve or Hitler, but with the difference that in each situation he is also ignorant of the right answers. No solution exists to the problem of how the side that lost a diplomatic negotiation or a battle might have won, even when one believes that fate *could* have gone the other way. But the historian of science always knows what the right answer is, and often finds it painfully easy to discover why it was not given in a faulty investigation (Hall, 1983, p. 56).

When historians of science emphasize the specificity of the historicity of science, they usually do so, just as Hall did, by arguing that science is an activity where historical change can be identified with progress (Mayr, 1990), at least when "a sequence of sufficiently large and distant events" is considered (Alvargonzàlez, 2013, p. 91). Progress in science must not be taken for granted as it can be observed in several of its dimensions: predictive capacity, explanatory power, empirical success, fruitfulness, etc. If science is generally progressive – and there is no good reason to doubt that – it means that recent concepts and explanations are, in the main, "better" than those developed in earlier periods. If that is correct, it is easy to understand and to define what I call *normative presentism*: *the use of present-day concepts and explanations to identify and emphasize the obstacles faced by earlier concepts in order to understand the same class of phenomena as well as their intrinsic limitations and theoretical shortcomings.*

For example, awareness of modern evolutionary theory allows us to understand that Lamarck's theory was ill equipped to convincingly address the fundamental problem of the apparent finality of the morphology and anatomy of living things (Lamarck, 1809), a problem later solved by Darwin when he elaborated the concept of adaptation by natural selection. The point is to be able to understand, better than Lamarck himself did, the theoretical tensions that characterized his work by being aware of obstacles that he was unable to make out – let alone overcome.

In my view, this is exactly the kind of presentism that Nick Tosh referred to when he concluded his 2003 article by writing that "the use of modern natural scientific theories in historical explanations cannot be ruled anachronistic *in the same way* as the use of modern social and political theories can" (Tosh, 2003, p. 658). *Normative*

[2] All translations of Canguilhem are mine. Existing English translations are not close enough to the French original for the purposes of the present article.

presentism was extensively discussed in France between the 1940s and the 1980s, mostly because of the impact of philosopher Gaston Bachelard's work. For Bachelard, whose main interest was in mathematical physics, science must be defined by its ability to progress (Bachelard, 1938 [1999], p. 17; Bachelard, 1951 [2008]; Bachelard, 1951, pp. 35–39; see also Section 4) and therefore the main – if not only – function of history of science is to value the present by showing its superiority in comparison to the past (Bachelard, 1951 [2008], pp. 160–162; Chimisso, 2015, p. 66). It is of course this philosophical definition of science that has been attacked since the 1970s by sociologists of knowledge and by the supporters of cultural relativism (Bontems, 2010, pp. 199–201), such as David Bloor (Bloor, 1976), Bruno Latour or Dominique Pestre (Pestre, 1995, pp. 489–490).

Since then, historicism and relativism have developed as alternatives in the field of history of science, albeit unsatisfactory ones because of their wholesale rejection of all forms of presentism. In the next section, I aim to characterize the limitations in the use of the forms of presentism outlined here. Being aware of them is the only way to rule out anachronism, Whiggism and positivism without giving up on presentism altogether.

3. Misuses of presentism: defining anachronism, positivism and Whigghism

Now that we have an operational typology of the most common and most essential forms of presentism, we can characterize in greater detail the three main types of bad practice in history of science stemming from a misuse of three of the four forms of presentism, namely *descriptive presentism*, *causal-narrative presentism* and *normative presentism*.

3.1. Anachronism as a misuse of descriptive presentism

The first type of bad practice is classically labelled "anachronism" and can be rigorously defined in light of *descriptive presentism*. The use of *descriptive presentism* can lead to the most typical dead end in the practice of history: to misrepresent or even ignore the intentions of past actors. In such cases, descriptions of past science can be incorrect in two ways.

First, as Yves Gingras already emphasized, they can be unfaithful because of the risk of conflating the actors' own categories and present categories (Gingras, 2001, p. 387). If we are allowed to use present concepts to provide a better understanding of the past, it is of course a mistake to suggest that these concepts were available to the actors, that they were part of their "repertoire" (Gingras, 2001, p. 409).

The second form of anachronism observed in history of science results from the apparent continuity of words and meanings. In describing past explanations we must be careful not to assume that the same words have the same meanings in different times. The case of the concept of virus and its changing meanings throughout the twentieth century is a textbook example of this. (Van Helvoort, 1991). It was first synonymous of microbe, i.e. an invisible, living entity able to parasitize cells. Then, during the 1950s, the concept of virus was reinterpreted in the wake of the development of molecular biology. It came to refer to a specific molecular system, instead of a living organism. Lack of awareness of this pivotal shift can cause major misinterpretations.

Anachronism thus arose as a misuse of *descriptive presentism*. When a historian describes the science of the past using present terms (analytical and/or scientific), he must be perfectly aware that some of these terms – or some of our understandings of these terms – were not known by the actors. In other words, anachronism is best defined as a lack of distinction between past actors' descriptions and the historian's analytical and scientific descriptions of the past.

3.2. Positivism and Whiggism as a misuse of causal-narrative and normative presentism

While the terms "Whiggism" and "positivism" are commonly used to denounce certain practices in history of science, they are often left undefined. In general, they are both quite interchangeably used to refer to inaccurate treatments of history, when historicity and contingency are clearly not taken into account. My aim here is not to draw a sharp distinction between Whigghism and positivism but rather to define them together. Whigghism might be seen as the strongest and resulting form of positivist history, but the difference between the content of these categories seems to me more quantitative then qualitative. Consequently, I will use these terms as synonyms.

Using the typology laid out in Section 2, it is now possible to propose two closely related definitions of these terms, each referring to one specific form of presentism between *causal-narrative presentism* and *normative presentism*. These two definitions rigorously encapsulate the conflated meanings of Whigghism seen in the existing literature.

Whigghism can first be a misuse of *causal-narrative presentism* because it equates causality with finality. The function of *causal-narrative presentism* is to identify causal series between historical periods. It directly questions the possibility and robustness of these historical lines. From a positivist standpoint – this was already the case in Auguste Comte's system – these links are absolute: scientific truth is a necessity which inevitably develops following a regular, predefined pattern (Radick, 2005, pp. 24–25). In this sense positivist history is rooted in the most simplistic and excessive use of *causal-narrative presentism*. This position is now called "inevitabilist" (Soler, Trizio, & Pickering, 2015). Inevitabilism posits that scientific truth has been there from the beginning, and history is only its necessary progressive discovery. Positivist history reduced historical time to a simple quantity, devoid of thickness. In such a case, Whigghism transforms *causal-narrative presentism* into some sort of *teleo-narrative presentism*: present science is not only the consequence of past science but also its purpose. This is precisely the concept of Whigghism that Butterfield opposed in his famous 1931 book, i.e. an over-linear and finalistic conception of history, where the past is "the ratification if not the glorification of the present" (Butterfield, 1931, p. 2).

But Whigghism can also refer to another kind of practice, closely related to the latter but nonetheless different. In this second case, it is a misuse of *normative presentism*, which can be observed quite frequently in the history of science written by scientists. Scientists are especially tempted to confer the status of a necessary norm to the present. Under the assumption that current science is *absolute* truth, it seems legitimate to use it as the ultimate standard to evaluate the past. But this kind of evaluation can quickly become a devaluation: science of the past is judged only to the extent that it fits with present science. As it is known (Chimisso, 2015, p. 65), Canguilhem himself was aware of this danger and remained very careful in the use of *normative presentism*[3]:

We can see that there is a big difference between recurrence, understood as a critical jurisdiction on the anteriority of a scientific present – bound to become outdated or altered, precisely because

[3] Note that Canguilhem's methodological stance against a certain (ab)use of *normative presentism* also suggests that his understanding of historical epistemology differed from Bachelard's. In the next section, I shall clarify this contrast by reactivating Canguilhem's project.

it is scientific – and the systematic and virtually mechanic application of a standard model of scientific theory operating as a sort of epistemological police of past theories. […] In that sense the work of the historian, equipped with a completed type of theory, would consist in probing past theories to identify the reasons of their lack of logical maturity. Retroactively applying a definitive current model considered as a universal touchstone is not selectively shedding light on the past; it is a form of historical blindness (Canguilhem, 2000 [1976], pp. 21–22).

The simplistic use of *normative presentism* implies that past explanations have value only on the basis of their resemblance to present explanation (i.e. the "truth"). If we come back to the example of Lamarck's and Darwin's evolutionary theories, then a positivist use of *normative presentism* would roughly consist in blaming Lamarck for not inventing natural selection in 1809. Lamarck is thus reduced to a precursor (Perrier, 1893), a scientist unable to complete the research program the historian forces him to be involved in. In history of science, this particular bias was recognized a long time ago, when historians criticized the propensity of scientists to invent precursors to their work (see for instance Canguilhem, 2002 [1968], pp. 20–21). Fortunately, this "*precursitis* disease", as Joseph T. Clark called it (Clark, 1969, p. 104), is nowadays well identified and might no longer represent a danger in the practice of history of science.

To sum up, Whigghism (or positivism) can be precisely defined as a two-fold misuse of presentism. It can be first an inappropriate understanding of the *historical genesis* of science. In this sense, Whigghism reduces history to a linear, uncontextualized (internalist) and finalistic series of events. But Whigghism can also be a simplistic normative judgement on the *content* of past science. In this sense, Whigghism devaluates past explanations on the grounds that they are not yet present truth.

Because of these misuses, historicism developed as an opposition to any use of presentism. In its stronger sense, historicism promoted the view that present knowledge is not a norm, but only a point of view (history of science is thus reduced to a kind of history of beliefs). At the core of this conception, at least as a methodological principle, one finds the postulate of equal validity of all forms of knowledge (between periods, between cultures, between science and non-science (Boghossian, 2006)). Present science is not seen as the best standard to evaluate the science of the past. That is why historicism rejects *normative presentism* and only allows the increasingly detailed description of scientific controversies (Collins, 2004; Pinch, 1986). Given this methodological positioning, neither the issue of truth nor that of rationality can be properly addressed. In the following section my aim is to show that historical epistemology acknowledges both the historicity *and* the rationality of science, and as such can help us overcome the limitations of Whiggism/positivism and historicism alike.

4. For a critical presentism: the project of historical epistemology

The research program of historical epistemology has clearly not been sufficiently defined, which has caused misunderstandings. For example, in their attempt to characterize historical epistemology, Thomas Sturm and Uljana Feest (Sturm & Feest, 2011, p. 285) call it (1) "a study of the history of higher-order epistemic concepts such as objectivity, observation, experimentation, or probability"; (2) "a study of the historical trajectories of the objects of research, such as the electron, DNA, or phlogiston"; (3) "a long-term study of scientific developments". This very inclusive characterization is the direct consequence of an unspecified use of the label "historical epistemology" by authors like Lorraine Daston or Jürgen Renn, which greatly differs from its use in Bachelard or Canguilhem (Chimisso, 2003, p. 298). This lack of definition fuels the arguments of critics who see the label "historical epistemology" – as used by Daston, Renn and others – as only a brand (Gingras, 2010).

Based on the clarification made in Section 2 about the variety of forms of presentism applied to history of science, I propose here to start rethinking the core of historical epistemology. The succinct presentation made in this section should not be seen as a definitive characterization of historical epistemology (and of what I call *critical presentism*), but only as a first step into that direction. Even if my aim here is primarily normative and not descriptive, my argument also draws on some pivotal statements made by Georges Canguilhem. In my view, it is Canguilhem, much more than Bachelard or Foucault, who best engaged with historical epistemology as a method for the historian of science. Bachelard did not always avoid the traps of positivism while Foucault clearly gave up all efforts towards normativity in his project of an archaeology (Chimisso, 2003, p. 300).

To begin with, a quick overview of the concept of progress in Bachelard's epistemology is necessary, because we shall see that Canguilhem deliberately distanced himself from it. In Bachelard's view, as it is well known, progress is an intrinsic and unavoidable dimension of the temporality of science. This allows him to state, in his famous text "On current developments in the history of sciences", that either the history of science "relates a growth, or it has nothing to say"[4] (Bachelard, 1951 [2008], p. 163). Scientific progress is nonetheless not linear: it is dialectical and discontinuous, involving epistemological breaks (Bachelard, 1928 [1973], p. 270; 1953, conclusion). For Bachelard, it is the very existence of such breaks that not only allows but requires us to use *normative presentism*, the core component of his epistemology. The normative power of present science is everywhere in his writings (Bachelard, 1938 [1999], p. 17; 1951, chapter 1). As a consequence, there is a sharp imbalance in Bachelard's system between retrospective judgment and chronological understanding, to the detriment of the latter. This imbalance ultimately comes from the fact that Bachelard remained a philosopher and not a historian of science (Gayon, 2003).

With Canguilhem, the center of gravity moves from the philosophy of science to the history of science. In a recent and illuminating in-depth study of Canguilhem's concept of scientific ideology, Cristina Chimisso evidences a tension between Canguilhem's project of constructing historical narratives and Bachelard's emphasis on the discontinuity between science and non-science (Chimisso, 2015). What is of interest for the present study is that Canguilhem, unlike Foucault, never repudiated the normative view of science that Bachelard promoted (Chimisso, 2015, p. 70). According to Chimisso's highly convincing argument, this is precisely why Canguilhem had to elaborate the concept of scientific ideology. It is also known that Canguilhem favored the term "epistemological history" over "historical epistemology" (Gayon, 2003), which reflected his greater interest in history. In the following paragraphs, my aim is to build on Chimisso's conclusions in making explicit how Canguilhem tried to bring together the historicity and the rationality of scientific knowledge (Chimisso, 2003, p. 300).

Two of Canguilhem's most famous texts are his late publications of 1968 and 1976 where he expounded at length on his conception of the interrelation between history of science and epistemology (Canguilhem, 2002 [1968]; Canguilhem, 2000 [1976]). Despite their importance and significance, it is not in those articles, however, that Canguilhem was the most clear about the issue at stake here. And because this problem proved to be central, Canguilhem

[4] My translation.

grappled with it in his first works involving at least to some extent the practice of history of science: *Essai sur quelques problèmes concernant le normal et le pathologique* (1943), *La théorie cellulaire* (1946), and *La formation du concept de réflexe aux XVIIe et XVIIIe siècles* (1955).

For the sake of brevity, I will now mostly rely on his long article about cell theory, written when he was a researcher at the University of Strasbourg at the end of World War II. Based on extensive and original studies of his colleague and friend Marc Klein (Klein, 1936; Limoges, 2015, pp. 32–33), Canguilhem published in 1946 a text first entitled "Cell theory in biology: On the sense and value of scientific theories" (Canguilhem, 2003 [1946]). The introduction features a discussion of Canguilhem's conception of the function of history of science.[5] In a few paragraphs, sometimes in allusive terms, Canguilhem outlines the core components of his practice as a historian of science, defining the project of what would later be called historical epistemology.

First, against positivism, he distances himself with the notion of progress as it was promoted in the Enlightenment and later in Auguste Comte's philosophy. Here, he opposes in very strong terms the over-simplistic use of *normative presentism* because – and, to my view, this is a serious matter of disagreement between Bachelard and him – the rule which holds that "chronological anteriority is logical inferiority" is wrong (Canguilhem, 2003 [1946], p. 54). As Camille Limoges has already pointed out (Limoges, 2015, p. 41), the contrast between Bachelard's and Canguilhem's understanding of *normative presentism* is particularly obvious if one compares Bachelard's *La formation de l'esprit scientifique* (1938) with Canguilhem's *La formation du concept de réflexe* (1955). On several occasions, Canguilhem explained why a dogmatic and positivist use of *normative presentism* could lead to a complete misinterpretation of history:

When we deem a scientific proposal to be true, we confer on it a retroactive validity that immediately exempts it from the fate of dreams, sketches, failures, mistakes – thoughts whose responsibility lies with someone. The elimination of falsehood by truth – i.e., verified fact – when achieved, appears as the almost mechanical effect of an irresistibly manifested impersonal necessity. This is why the importation of these judgment standards in the field of history can only create misunderstandings. When we review the work of various researchers in a scientific genealogy – which only specialists are competent enough to do – we continue to obey the rule of the retroactive effect of truth, as we retrace the history of a question according to the logic of the truth of the day, identified with the truth of always (Canguilhem, 1955, p. 156, p. 156).

Thus, for Canguilhem, *normative presentism* does not have the same driving function that it has for Bachelard, the theoretician of the recurrence method (Bachelard, 1951, chapter 1; Braunstein, 2008a, p. 101). Where Bachelard supports the intrinsic necessity of progress in science, Canguilhem conceives it rather as a distant perspective, based on a more statistical understanding (it must be recalled, here, that Canguilhem was a specialist of medicine and biology, sciences which have not demonstrated the same form of progress as mathematical physics (Chimisso, 2015, p. 71)). In Canguilhem's view, *normative presentism* is more regulatory than directive: it marks out the space within which historical investigation has to be deployed. *Normative presentism* is thus also used as a railing protecting historians of science from the temptation of erudition (Canguilhem, 1955, p. 158; later Canguilhem will oppose the idea that history is mainly the memory of the past (Canguilhem, 2000 [1976], p. 12)).

A few paragraphs later in his 1946 text, Canguilhem tackles the issue of historicity and contingency in the history of science in typically dense fashion:

A scientific work is rewarding if it does not impose the methodological or doctrinal choice that it leans towards. The reasons for that choice must be sought elsewhere than in the work. In my view the benefit of a well-conceived history of science is to reveal history in science. By history, I mean the sense of possibility. Knowledge is not so much about struggling with reality as about validating a possible outcome by making it necessary. In that sense the genesis of what is possible matters just as much as the demonstration of what is necessary. The former's frailty does not deprive it of a dignity that the latter might derive from its solidity. Illusion could have been truth. Truth might some day turn out to be illusion. (Canguilhem, 2003 [1946], p. 58).

First, Canguilhem is very clear about the historicity of history of science: it is real and contingency must be considered as a dimension of history of science. History of science is therefore true history, and has to show the "sense of possibility", i.e. the contingent modalities of the "genesis of what is possible". It is however in the second part of the paragraph that Canguilhem is the most original yet also the most elliptic. If history of science must deal with historicity (science could have been different), in contrast to what happens in history, history of science must also take into account a specific kind of "necessity": history of science is expected to explain the "demonstration of what is necessary". What did he mean? "Making possibility necessary", in my reading, is the acknowledgement of the fact that scientific knowledge has a transhistorical dimension as soon as its rationality has been successfully tested. "Validating a possible outcome" thus consists, without contradiction, in making the contingent rational and thus necessary. If science could have been different, this does not mean at all that there is no demarcation between science and other human discourses and practices. For Canguilhem, science is at the same time contingent *and* rational. But the rationality of scientific knowledge can only be observed in the "longue durée", in the course of the progressive, difficult, and tortuous demonstration of the necessity of this knowledge.[6]

The history of the concept of natural selection perfectly illustrates this crucial aspect of the project of historical epistemology (Gayon, 1998). Its elaboration was partly contingent, because Charles Darwin's intentions, for example, were quite different from those of Alfred Russel Wallace. Darwin thus steered the history of the concept of natural selection in a specific direction (to such an extent that contemporary debates about the levels of selection have to come back to Darwin's own writings (Gould, 2002)). Yet it isn't Darwin who fully proved the rationality of natural selection. It is well known that biologists only completed the theoretical framework in which this concept became operational in the late 1930s and the 1940s. Natural selection was rationalized only in population genetics (Gayon, 1998). Population genetics thus made necessary what was previously considered by most biologists as a complete dead-end (Bowler, 1992).

Therefore, as he considers the contingent and the rational to be inseparable, Canguilhem opposes both the over-simplistic use of

[5] This introduction is missing from the English translation included in the 1994 vol edited by François Delaporte (Delaporte, 1994).

[6] While this distinction between the contingent birth of a given knowledge and its subsequent rationalization might appear rather close to Hans Reichenbach's famous distinction between the context of discovery and the context of justification (Reichenbach, 1938), this is not the case at all. In Reichenbach's work, only the path towards a "discovery" is contingent, not its content, whereas in the case of historical epistemology, it is excessive to speak of a discovery, because science could have been different.

causal-narrative and *normative presentisms* of positivist history and the anti-presentist relativism of the historicist approach (Canguilhem, 2002 [1968], p. 15). Canguilhem tentatively uses *causal-narrative presentism* to indicate the fact that the historical sequence could have been different. In that sense, his methodological positioning is quite close to those who support counterfactual history and methodological contingentism (Soler, 2015, p. 42): against positivism, scientific truth is a transient historical accident. But Canguilhem still makes room for *normative presentism*. In opposition to historicism, he nonetheless thinks of the science of the present as a norm, and the historian also has to reactivate its progressive rationalization. For Canguilhem the present in science is akin to a contingent norm. Because it is a norm, it legitimates a careful *normative presentism*. But because this norm could have been different, it also requires the non-finalistic use of *causal-narrative presentism*. The historian of science should never forget that "illusion could have been truth [and] truth might some day turn out to be illusion" (the "truth of the day" is not the "truth of always" as he insisted again in 1955 (p. 156)).

Historicism pictures scientific truth as an accident frozen in time for sociological reasons whereas positivism understands it as an inescapable necessity. A third way is provided by historical epistemology, whose research program is based – or should be based – on the idea that scientific truth is an accident made to some extent necessary. This counter-intuitive intricacy of contingency and rationality allows this research program to overcome the dead-ends of Whiggism and relativism.

Moreover, the relations between the present and the past must be conceived as reciprocal. If the present is the norm from which it is possible to perform careful recurrences towards the past (*normative presentism*), the fact that this understanding works in accordance with the idea of the contingency of historical sequences (*causal-narrative presentism*) allows the reverse perspective: by reactivating the complexity of the past, one develops tools to criticize present science. Awareness of the fleetingness of the present is the best response to scientific dogmatism. This essential critical function of historical epistemology has been suggested since the beginning by Canguilhem, who, following Paul Langevin, thought that "to groom new minds for scientific work, i.e., […] for calling into question certain solutions, returning to the sources is indispensable" (Canguilhem, 2003 [1946], p. 56).

This specific form of presentism, which I call *critical presentism*, could be therefore defined as *the elaboration of a reciprocal critical relation between scientific knowledges from different times*. Its legitimacy comes from the acknowledgement of both the contingency and the rationality of science. This *critical presentism*, which can be understood as a way to move past *causal-narrative* and *normative* presentism, precisely defines the project of historical epistemology.

Critical presentism has already proven to be important in several occasions. In the course of history of science, there have been scientists who were able to bring substantial insights into their field of research and to develop an extensive historical competence. More interestingly, their historical knowledge was most of the time one of the main reasons for their scientific achievements. The most famous case here might be that of Austrian physicist Ernst Mach. One of his main books, *Die Mechanik in ihrer Entwicklung. Historish-kritisch dargestellt* (*The Science of Mechanics)*, first published in 1883, was a philosophical and historical analysis of the development of mechanics that famously played a key part in the development of the new theory of relativity (Blackmore, 1972). Closer to us, the American paleontologist Stephen Jay Gould, in collaboration with Niles Eldredge, developed a new understanding of the process of macroevolution – namely the theory of "punctuated equilibria" (Eldredge & Gould, 1972) – and published an impressive body of work as a historian of the life sciences. His last scientific book, *The Structure of Evolutionary Theory*, is a fascinating object at the intersection of science, philosophy and history (Gould, 2002). According to Gould himself, the history of biology was always a fundamental part in his quest to reform evolutionary theory (Gould, 1977, pp. 5–6). As such, Mach and Gould were doing historical epistemology in the sense defined above.

At least on one famous occasion, Canguilhem himself tried to have a critical and transformative action on present science. His medical PhD thesis *Le normal et le pathologique* (*The Normal and the Pathological*) is an explicit attempt to show that these two categories are not well conceived in present-day medicine because the pathological state is reduced to only a quantitative modification of the normal state (Canguilhem, 2003 [1966]). Canguilhem partly based his demonstration on an examination of the formation of these concepts in the history of medicine, from Broussais and Claude Bernard to the physiology of the 1930s and 1940s. But his ambition was not historical, it was to "contribute to the renewal of certain methodological concepts by rectifying their understanding' (Canguilhem, 2003 [1966], p. 8). Canguilhem argued that philosophy must "reconsider issues rather than settle them" (Canguilhem, 2003 [1966], p. 9). This critical project always remained pivotal in his understanding of the practice of historical epistemology.

5. Conclusion

My aim in this study was two-fold. First, I have tried to lay out a typology of what I consider being the four major forms of presentism at stake in the field of history of science, which I call *empirical, descriptive, causal-narrative* and *normative presentism*. In my view, all of these forms could be legitimate and useful, even if this might require some epistemological dexterity. That Whiggism and positivism are not satisfactory means to produce narratives does not mean we should give up on presentism altogether. The fourth form of presentism – *normative presentism* – is specific to history of science: history of science differs from general history due to the *trans*-historical dimension of its object which involves a complex relation to the present. A strictly historicist perspective is thus impossible because it would mean giving up understanding what is consubstantial with science, i.e. its necessary connection with progress and truth. On the contrary, positivism and Whiggism are based over-simplistic and over-dogmatic uses of both *causal-narrative* and *normative presentisms*. They imply the refusal of the historicity and contingency of science. Present science is then pictured as a necessary norm and past science as only a transient step leading to it.

Secondly, this conceptual clarification of the various forms of presentism allows for a general characterization of the project of historical epistemology, which deserves to be revived. Historical epistemology, at least if we expand Georges Canguilhem's approach, contrasts (or should contrast) with both Whiggism/positivism on the one hand and historicism on the other. Its project is defined by a fifth form of presentism which I call *critical presentism*. This resulting form acknowledges the contingent *and* rational dimensions of scientific knowledge. Because science progresses, then it is legitimate for the present to serve as a yardstick for judging the past. But because the present is at the same time a contingent and fleeting norm, then the past becomes the privileged tool for criticizing the dogmatist temptation of present science. This succinct presentation of *critical presentism* is only a first step in the process of rethinking what historical epistemology should be.

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